

РБНФ №1 (опис синтаксису всіма допустимими засобами РБНФ)	РБНФ №2 (опис формальної граматики засобами РБНФ)	Формальна граматика	Формальна граматика з специфікацією lookahead у правилах для LL(2)-аналізатора	Перевірка РБНФ №1 за допомогою коду <i>(помістити у файл "EBNF_N1.h")</i> */	Перевірка РБНФ №2 за допомогою коду <i>(помістити у файл "EBNF_N2.h")</i> */	Перевірки прототипу LL(2)-синтаксичного аналізатора (спеціальна структура) та прототипу лексичного аналізатора (регулярні вирази) за допомогою коду. Лексеми для синтаксичного аналізатора обробляються лексичним аналізатором, тому синтаксичний аналізатор не аналізує їх повторно (як показано в РБНФ). <i>(помістити у файл "LexicaByRegExAndSyntaxByLL2prototype.h")</i> УВАГА: при копіюванні зважайте, щоб у кожному рядку після символу «\» не містилось жодних інших символів. */
		G = {N, T, P, S}	G = {N, T, P, S}			
		S → program_rule	S → program_rule			
		N = { program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false_optional, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, statements_or_block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	N = { program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false_optional, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, statements_or_block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	#define NONTERMINALS program_name, \ value_type, \ array_specify, \ declaration_element, \ \ other_declaration_ident, \ declaration, \ \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ \ expression, \ \ expression_or_cond_block__with_optional_assign, \ assign_to_right, \ \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ \ \ continue_while, \ break_while, \ statement_in_while_and_if_body, \ statement, \ block_statements_in_while_and_if_body, \ \ while_cycle_head_expression, \ while_cycle, \ statements_or_block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ \ \ program_rule, \ \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	#define NONTERMINALS program_name, \ value_type, \ array_specify, \ declaration_element, \ array_specify_optional, \ other_declaration_ident, \ declaration, \ other_declaration_ident__iteration, \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ index_action_optional, \ expression, \ binary_action__iteration, \ expression_or_cond_block__with_optional_assign, \ assign_to_right, \ assign_to_right_optional, \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ false_cond_block_without_else__iteration, \ body_for_false_optional, \ continue_while, \ break_while, \ statement_in_while_and_if_body, \ statement, \ block_statements_in_while_and_if_body, \ statement_in_while_and_if_body__iteration, \ while_cycle_head_expression, \ while_cycle, \ statements_or_block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ statement__iteration, \ expression_optional, \ program_rule, \ declaration_optional, \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ sign_optional, \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	
		"INTEGER16", " ,", "NOT", "AND", "OR", "==" , "!=", "<=", ">=", "+", " ,", " ,", "@" , "DIV", "MOD", "(", 	"INTEGER16", " ,", "NOT", "AND", "OR", "==" , "!=", "<=", ">=", "+", " ,", " ,", "@" , "DIV", "MOD", "(", 	#define TOKENS \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \	#define TOKENS \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \	

		<pre>"")", "=", "ELSE", "IF", "WHILE", "CONTINUE", "BREAK", "EXIT", "GET", "PUT", "NAME", "BODY", "DATA", "BEGIN", "END", "(", ")", "[", "]", "(", ")", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", " ", "A", "B", "C", "D", "E", "F", "G", "H", "I", "J", "K", "L", "M", "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z", "a", "b", "c", "d", "e", "f", "g", "h", "i", "j", "k", "l", "m", "n", "o", "p", "q", "r", "s", "t", "u", "v", "w", "x", "y", "z"} </pre>	<pre>"")", "=", "ELSE", "IF", "WHILE", "CONTINUE", "BREAK", "EXIT", "GET", "PUT", "NAME", "BODY", "DATA", "BEGIN", "END", "(", ")", "[", "]", "(", ")", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", " ", "A", "B", "C", "D", "E", "F", "G", "H", "I", "J", "K", "L", "M", "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z", "a", "b", "c", "d", "e", "f", "g", "h", "i", "j", "k", "l", "m", "n", "o", "p", "q", "r", "s", "t", "u", "v", "w", "x", "y", "z"} </pre>	<pre>tokenGROUPEXPRESSIONEND, \ tokenLRASSIGN, \ tokenELSE, \ tokenIF, \ tokenWHILE, \ tokenCONTINUE, \ tokenBREAK, \ tokenEXIT, \ tokenGET, \ tokenPUT, \ tokenNAME, \ tokenBODY, \ tokenDATA, \ tokenBEGIN, \ tokenEND, \ tokenBEGINBLOCK, \ tokenENDBLOCK, \ tokenLEFTSQUAREBRACKETS, \ tokenRIGHTSQUAREBRACKETS, \ tokenSEMICOLON, \ digit_0, \ digit_1, \ digit_2, \ digit_3, \ digit_4, \ digit_5, \ digit_6, \ digit_7, \ digit_8, \ digit_9, \ tokenUNDERSCORE, \ A, \ B, \ C, \ D, \ E, \ F, \ G, \ H, \ I, \ J, \ K, \ L, \ M, \ N, \ O, \ P, \ Q, \ R, \ S, \ T, \ U, \ V, \ W, \ X, \ Y, \ Z, \ a, \ b, \ c, \ d, \ e, \ f, \ g, \ h, \ i, \ j, \ k, \ l, \ m, \ n, \ o, \ p, \ q, \ r, \ s, \ t, \ u, \ v, \ w, \ x, \ y, \ z </pre>	<pre>tokenGROUPEXPRESSIONEND, \ tokenLRASSIGN, \ tokenELSE, \ tokenIF, \ tokenWHILE, \ tokenCONTINUE, \ tokenBREAK, \ tokenEXIT, \ tokenGET, \ tokenPUT, \ tokenNAME, \ tokenBODY, \ tokenDATA, \ tokenBEGIN, \ tokenEND, \ tokenBEGINBLOCK, \ tokenENDBLOCK, \ tokenLEFTSQUAREBRACKETS, \ tokenRIGHTSQUAREBRACKETS, \ tokenSEMICOLON, \ digit_0, \ digit_1, \ digit_2, \ digit_3, \ digit_4, \ digit_5, \ digit_6, \ digit_7, \ digit_8, \ digit_9, \ tokenUNDERSCORE, \ A, \ B, \ C, \ D, \ E, \ F, \ G, \ H, \ I, \ J, \ K, \ L, \ M, \ N, \ O, \ P, \ Q, \ R, \ S, \ T, \ U, \ V, \ W, \ X, \ Y, \ Z, \ a, \ b, \ c, \ d, \ e, \ f, \ g, \ h, \ i, \ j, \ k, \ l, \ m, \ n, \ o, \ p, \ q, \ r, \ s, \ t, \ u, \ v, \ w, \ x, \ y, \ z </pre>	<pre>#define COMMENT_BEGIN_STR "##" #define COMMENT_END_STR "##" #define TOKENS_RE "[:> = { ~ + - * . , . = { } \\[\\] \\(\\) < > = _ _0-9A-Za-z ([^\\"\\V\\F\\v\\n])" #define IDENTIFIERS_RE "_[A-Z][A-Z][A-Z][A-Z][A-Z][A-Z]" #define UNSIGNEDVALUES_RE "0 [1-9][0-9]+"</pre>	<pre>#define T_BEGIN_GROUPEXPRESSION_0 "(" #define T_BEGIN_GROUPEXPRESSION_1 "" #define T_BEGIN_GROUPEXPRESSION_2 "" #define T_BEGIN_GROUPEXPRESSION_3 "" #define T_END_GROUPEXPRESSION_0 ")" #define T_END_GROUPEXPRESSION_1 "" #define T_END_GROUPEXPRESSION_2 "" #define T_END_GROUPEXPRESSION_3 "" #define T_LEFT_SQUAREBRACKETS_0 "[" #define T_LEFT_SQUAREBRACKETS_1 "" #define T_LEFT_SQUAREBRACKETS_2 "" #define T_LEFT_SQUAREBRACKETS_3 "" #define T_RIGHT_SQUAREBRACKETS_0 "]" #define T_RIGHT_SQUAREBRACKETS_1 "" #define T_RIGHT_SQUAREBRACKETS_2 "" #define T_RIGHT_SQUAREBRACKETS_3 "" </pre>
				tokenGROUPEXPRESSIONBEGIN = "(" >> BOUNDARIES;	tokenGROUPEXPRESSIONBEGIN = "(" >> BOUNDARIES;		
				tokenGROUPEXPRESSIONEND = ")" >> BOUNDARIES;	tokenGROUPEXPRESSIONEND = ")" >> BOUNDARIES;		
				tokenLEFTSQUAREBRACKETS = "[" >> BOUNDARIES;	tokenLEFTSQUAREBRACKETS = "[" >> BOUNDARIES;		
				tokenRIGHTSQUAREBRACKETS = "]" >> BOUNDARIES;	tokenRIGHTSQUAREBRACKETS = "]" >> BOUNDARIES;		

				tokenBEGINBLOCK = "{" >> BOUNDARIES;	tokenBEGINBLOCK = "{" >> BOUNDARIES;	#define T_BEGIN_BLOCK_0 "{" #define T_BEGIN_BLOCK_1 "" #define T_BEGIN_BLOCK_2 "" #define T_BEGIN_BLOCK_3 ""	
				tokenENDBLOCK = "}" >> BOUNDARIES;	tokenENDBLOCK = "}" >> BOUNDARIES;	#define T_END_BLOCK_0 "}" #define T_END_BLOCK_1 "" #define T_END_BLOCK_2 "" #define T_END_BLOCK_3 ""	
				tokenSEMICOLON = ";" >> BOUNDARIES;	tokenSEMICOLON = ";" >> BOUNDARIES;	#define T_SEMICOLON_0 ";" #define T_SEMICOLON_1 "" #define T_SEMICOLON_2 "" #define T_SEMICOLON_3 ""	
				tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	#define T_DATA_TYPE_0 "INTEGER16" #define T_DATA_TYPE_1 "" #define T_DATA_TYPE_2 "" #define T_DATA_TYPE_3 ""	
				tokenCOMMA = "," >> BOUNDARIES;	tokenCOMMA = "," >> BOUNDARIES;	#define T_COMA_0 "," #define T_COMA_1 "" #define T_COMA_2 "" #define T_COMA_3 ""	
						#define T_BITWISE_NOT_0 "~" #define T_BITWISE_NOT_1 "" #define T_BITWISE_NOT_2 "" #define T_BITWISE_NOT_3 ""	
				tokenNOT = "NOT" >> STRICT_BOUNDARIES;	tokenNOT = "NOT" >> STRICT_BOUNDARIES;	#define T_NOT_0 "NOT" #define T_NOT_1 "" #define T_NOT_2 "" #define T_NOT_3 ""	
						#define T_BITWISE_AND_0 "&" #define T_BITWISE_AND_1 "" #define T_BITWISE_AND_2 "" #define T_BITWISE_AND_3 ""	
				tokenAND = "AND" >> STRICT_BOUNDARIES;	tokenAND = "AND" >> STRICT_BOUNDARIES;	#define T_AND_0 "AND" #define T_AND_1 "" #define T_AND_2 "" #define T_AND_3 ""	
						#define T_BITWISE_OR_0 " " #define T_BITWISE_OR_1 "" #define T_BITWISE_OR_2 "" #define T_BITWISE_OR_3 ""	
				tokenOR = "OR" >> STRICT_BOUNDARIES;	tokenOR = "OR" >> STRICT_BOUNDARIES;	#define T_OR_0 "OR" #define T_OR_1 "" #define T_OR_2 "" #define T_OR_3 ""	
				tokenEQUAL = "==" >> BOUNDARIES;	tokenEQUAL = "==" >> BOUNDARIES;	#define T_EQUAL_0 "==" #define T_EQUAL_1 "" #define T_EQUAL_2 "" #define T_EQUAL_3 ""	
				tokenNOTEQUAL = "!=" >> BOUNDARIES;	tokenNOTEQUAL = "!=" >> BOUNDARIES;	#define T_NOT_EQUAL_0 "!=" #define T_NOT_EQUAL_1 "" #define T_NOT_EQUAL_2 "" #define T_NOT_EQUAL_3 ""	
				tokenLESSOREQUAL = "<=" >> BOUNDARIES;	tokenLESSOREQUAL = "<=" >> BOUNDARIES;	#define T_LESS_OR_EQUAL_0 "<="	#define T_LESS_OR_EQUAL_1 "" #define T_LESS_OR_EQUAL_2 "" #define T_LESS_OR_EQUAL_3 ""
				tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	#define T_GREATER_OR_EQUAL_0 ">="	#define T_GREATER_OR_EQUAL_1 "" #define T_GREATER_OR_EQUAL_2 "" #define T_GREATER_OR_EQUAL_3 ""
				tokenPLUS = "+" >> BOUNDARIES;	tokenPLUS = "+" >> BOUNDARIES;	#define T_ADD_0 "+" #define T_ADD_1 "" #define T_ADD_2 "" #define T_ADD_3 ""	
				tokenMINUS = "-" >> BOUNDARIES;	tokenMINUS = "-" >> BOUNDARIES;	#define T_SUB_0 "-" #define T_SUB_1 "" #define T_SUB_2 "" #define T_SUB_3 ""	
				tokenMUL = "*" >> BOUNDARIES;	tokenMUL = "*" >> BOUNDARIES;	#define T_MUL_0 "" #define T_MUL_1 "" #define T_MUL_2 "" #define T_MUL_3 ""	
				tokenDIV = "DIV" >> STRICT_BOUNDARIES;	tokenDIV = "DIV" >> STRICT_BOUNDARIES;	#define T_DIV_0 "DIV" #define T_DIV_1 "" #define T_DIV_2 "" #define T_DIV_3 ""	
				tokenMOD = "MOD" >> STRICT_BOUNDARIES;	tokenMOD = "MOD" >> STRICT_BOUNDARIES;	#define T_MOD_0 "MOD" #define T_MOD_1 "" #define T_MOD_2 "" #define T_MOD_3 ""	
				tokenLRASSIGN = "=:;" >> BOUNDARIES;	tokenLRASSIGN = "=:;" >> BOUNDARIES;	#define T_LRASSIGN_0 "=:;" #define T_LRASSIGN_1 "" #define T_LRASSIGN_2 "" #define T_LRASSIGN_3 ""	
						#define T_THEN_BLOCK_0 "{" #define T_THEN_BLOCK_1 "" #define T_THEN_BLOCK_2 "" #define T_THEN_BLOCK_3 ""	
				tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	#define T_ELSE_BLOCK_0 "ELSE" #define T_ELSE_BLOCK_1 T_BEGIN_BLOCK_0 #define T_ELSE_BLOCK_2 "" #define T_ELSE_BLOCK_3 ""	
				tokenIF = "IF" >> STRICT_BOUNDARIES;	tokenIF = "IF" >> STRICT_BOUNDARIES;	#define T_IF_0 "IF" #define T_IF_1 "" #define T_IF_2 "" #define T_IF_3 ""	
						#define T_ELSE_IF_0 "ELSE" #define T_ELSE_IF_1 T_IF_0 #define T_ELSE_IF_2 "" #define T_ELSE_IF_3 ""	
				tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	#define T_WHILE_0 "WHILE" #define T_WHILE_1 "" #define T_WHILE_2 "" #define T_WHILE_3 ""	
				tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	#define T_CONTINUE_WHILE_0 "CONTINUE" #define T_CONTINUE_WHILE_1 "" #define T_CONTINUE_WHILE_2 ""	

					#define T_CONTINUE_WHILE_3 ""
				tokenBREAK = "BREAK" >> STRICT_BOUNDARIES;	#define T_EXIT_WHILE_0 "BREAK" #define T_EXIT_WHILE_1 "" #define T_EXIT_WHILE_2 "" #define T_EXIT_WHILE_3 ""
				tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	#define T_EXIT_0 "EXIT" #define T_EXIT_1 "" #define T_EXIT_2 "" #define T_EXIT_3 ""
				tokenGET = "GET" >> STRICT_BOUNDARIES;	#define T_INPUT_0 "GET" #define T_INPUT_1 "" #define T_INPUT_2 "" #define T_INPUT_3 ""
				tokenPUT = "PUT" >> STRICT_BOUNDARIES;	#define T_OUTPUT_0 "PUT" #define T_OUTPUT_1 "" #define T_OUTPUT_2 "" #define T_OUTPUT_3 ""
				tokenNAME = "NAME" >> STRICT_BOUNDARIES;	#define T_NAME_0 "NAME" #define T_NAME_1 "" #define T_NAME_2 "" #define T_NAME_3 ""
				tokenBODY = "BODY" >> STRICT_BOUNDARIES;	#define T_BODY_0 "BODY" #define T_BODY_1 "" #define T_BODY_2 "" #define T_BODY_3 ""
				tokenDATA = "DATA" >> STRICT_BOUNDARIES;	#define T_DATA_0 "DATA" #define T_DATA_1 "" #define T_DATA_2 "" #define T_DATA_3 ""
				tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	#define T_BEGIN_0 "BEGIN" #define T_BEGIN_1 "" #define T_BEGIN_2 "" #define T_BEGIN_3 ""
				tokenEND = "END" >> STRICT_BOUNDARIES;	#define T_END_0 "END" #define T_END_1 "" #define T_END_2 "" #define T_END_3 ""
					#define T_NULL_STATEMENT_0 "NULL" #define T_NULL_STATEMENT_1 "STATEMENT" #define T_NULL_STATEMENT_2 "" #define T_NULL_STATEMENT_3 ""
					#define GRAMMAR_LL2_2025 {\
program_name = ident;	program_name = ident;	program_name → ident	program_name(1: "ident_terminal") → ident	program_name = SAME_RULE(ident);	{ LA_IS, ("ident_terminal"), { "program_name", {\
value_type = "INTEGER16";	value_type = "INTEGER16";	value_type → "INTEGER16"	value_type(1: "INTEGER16") → "INTEGER16"	value_type = SAME_RULE(tokenINTEGER16);	{ LA_IS, (T_DATA_TYPE_0), { "value_type", {\
	array_specify = "[", unsigned_value, "]"	array_specify → "[" unsigned_value "]"	array_specify(1: "["") → "[" unsigned_value "]"		{ LA_IS, ("["), { "array_specify", {\
declaration_element = ident, array_specify__optional;	declaration_element = ident, array_specify__optional;	declaration_element → ident array_specify__optional	declaration_element(1: "ident_terminal") → ident array_specify__optional	declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);	{ LA_IS, ("ident_terminal"), { "declaration_element", {\
	array_specify__optional = array_specify ε;	array_specify__optional → array_specify array_specify__optional → ε	array_specify__optional(1: "["") → array_specify array_specify__optional(1: !"["") → ε		{ LA_IS, ("["), { "array_specify__optional", {\
other_declaration_ident = ",", declaration_element;	other_declaration_ident = ",", declaration_element;	other_declaration_ident → ",", declaration_element	other_declaration_ident(1: ",") → ",", declaration_element	other_declaration_ident = tokenCOMMA >> declaration_element;	{ LA_IS, (T_COMA_0), { "other_declaration_ident", {\
declaration = value_type, declaration_element, other_declaration_ident_iteration;	declaration = value_type, declaration_element, other_declaration_ident_iteration;	declaration → value_type declaration_element other_declaration_ident_iteration	declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident_iteration	declaration = value_type >> declaration_element >> *other_declaration_ident;	{ LA_IS, (T_DATA_TYPE_0), { "declaration", {\
	other_declaration_ident_iteration = other_declaration_ident, other_declaration_ident_iteration ε;	other_declaration_ident_iteration → other_declaration_ident other_declaration_ident_iteration false_cond_block_without_else_iteration → ε	other_declaration_ident_iteration(1: ",") → other_declaration_ident other_declaration_ident_iteration false_cond_block_without_else_iteration(1: !"") → ε		{ LA_IS, (T_COMA_0), {\
index_action = "[", expression, "]"	index_action = "[", expression, "]"	index_action → "[" expression "]"	index_action(1: "["") → "[" expression "]"	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	{ LA_IS, ("["), { "index_action", {\
unary_operator = "NOT";	unary_operator = "NOT";	unary_operator → "NOT"	unary_operator(1: "NOT") → "NOT"	unary_operator = SAME_RULE(tokenNOT);	{ LA_IS, (T_NOT_0), { "unary_operator", {\
unary_operation = unary_operator, expression;	unary_operation = unary_operator, expression;	unary_operation → unary_operator expression	unary_operation(1: "NOT") → unary_operator expression	unary_operation = unary_operator >> expression;	{ LA_IS, (T_NOT_0), { "unary_operation", {\
binary_operator = "AND" "OR" "==" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator = "AND" "OR" "==" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator → "AND" binary_operator → "OR" binary_operator → "==" binary_operator → "!=" binary_operator → "<=" binary_operator → ">=" binary_operator → "+" binary_operator → "-" binary_operator → "*" binary_operator → "DIV" binary_operator → "MOD"	binary_operator(1: "AND") → "AND" binary_operator(1: "OR") → "OR" binary_operator(1: "==") → "==" binary_operator(1: "!=") → "!=" binary_operator(1: "<=") → "<=" binary_operator(1: ">=") → ">=" binary_operator(1: "+") → "+" binary_operator(1: "-") → "-" binary_operator(1: "*") → "*" binary_operator(1: "DIV") → "DIV" binary_operator(1: "MOD") → "MOD"	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	{ LA_IS, (T_AND_0), { "binary_operator", {\

	body_for_false__optional;	body_for_false__optional	false_cond_block_without_else__iteration body_for_false__optional		false_cond_block_without_else__iteration >> body_for_false__optional;	"body_for_true", "false_cond_block_without_else__iteration", "body_for_false__optional" }}\n }}}}\n
	false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration ε;	false_cond_block_without_else__iteration → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration → ε	false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → ε false_cond_block_without_else__iteration(1: !"ELSE") → ε		false_cond_block_without_else__iteration = false_cond_block_without_else >> false_cond_block_without_else__iteration "";	{LA_IS, {T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration", {\n {LA_IS, {T_ELSE_IF_1 }, 2, { "false_cond_block_without_else", "false_cond_block_without_else__iteration" }}}, \n {LA_NOT, {T_ELSE_IF_1 }, 0, { "" }}\n }}}}\n {LA_NOT, {T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration", {\n {LA_IS, {"" }, 0, { "" }}\n }}}}\n
	body_for_false__optional = body_for_false ε;	body_for_false__optional → body_for_false body_for_false__optional → ε	body_for_false__optional(1: "FALSE") → body_for_false body_for_false__optional(1: !"FALSE") → ε		body_for_false__optional = body_for_false "";	{LA_IS, {T_ELSE_BLOCK_0 }, { "body_for_false__optional", {\n {LA_IS, {"" }, 1, { "body_for_false" }}\n }}}}\n {LA_NOT, {T_ELSE_BLOCK_0 }, { "body_for_false__optional", {\n {LA_IS, {"" }, 0, { "" }}\n }}}}\n
	continue_while = "CONTINUE";	continue_while → "CONTINUE"	continue_while(1: "CONTINUE") → "CONTINUE"	continue_while = SAME_RULE(tokenCONTINUE);	continue_while = SAME_RULE(tokenCONTINUE);	{LA_IS, {T_CONTINUE_WHILE_0 }, { "continue_while", {\n {LA_IS, {"" }, 1, { T_CONTINUE_WHILE_0 }}\n }}}}\n
	break_while = "BREAK";	break_while → "BREAK"	break_while(1: "BREAK") → "BREAK"	break_while = SAME_RULE(tokenBREAK);	break_while = SAME_RULE(tokenBREAK);	{LA_IS, {T_EXIT_WHILE_0 }, { "break_while", {\n {LA_IS, {"" }, 1, { T_EXIT_WHILE_0 }}\n }}}}\n
statement_in_while_and_if_body = statement "CONTINUE" "BREAK";	statement_in_while_and_if_body = statement continue_while break_while;	statement_in_while_and_if_body → statement statement_in_while_and_if_body → continue_while statement_in_while_and_if_body → break_while	statement_in_while_and_if_body(1: " ", "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "WHILE", "GET", "PUT", ",") → statement statement_in_while_and_if_body(1: "CONTINUE") → continue_while statement_in_while_and_if_body(1: "BREAK") → break_while	statement_in_while_and_if_body = statement continue_while break_while;	statement_in_while_and_if_body = statement continue_while break_while;	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement_in_while_and_if_body", {\n {LA_IS, {"" }, 1, { "statement" }}\n }}}}\n {LA_IS, {T_CONTINUE_WHILE_0 }, { "statement_in_while_and_if_body", {\n {LA_IS, {"" }, 1, { "continue_while" }}\n }}}}\n {LA_IS, {T_EXIT_WHILE_0 }, { "statement_in_while_and_if_body", {\n {LA_IS, {"" }, 1, { "break_while" }}\n }}}}\n
block_statements_in_while_and_if_body = "(" , {statement_in_while_and_if_body} , ")";	block_statements_in_while_and_if_body = "(" , statement_in_while_and_if_body__iteration , ")";	block_statements_in_while_and_if_body → "(" statement_in_while_and_if_body__iteration ")"	block_statements_in_while_and_if_body(1: "(") → "(" statement_in_while_and_if_body__iteration ")"	block_statements_in_while_and_if_body = tokenBEGINBLOCK >> *statement_in_while_and_if_body >> tokenENDBLOCK;	block_statements_in_while_and_if_body = tokenBEGINBLOCK >> statement_in_while_and_if_body__iteration >> tokenENDBLOCK;	{LA_IS, {T_BEGIN_BLOCK_0 }, { "block_statements_in_while_and_if_body", {\n {LA_IS, {"" }, 3, { T_BEGIN_BLOCK_0, "statement_in_while_and_if_body__iteration", T_END_BLOCK_0 }}\n }}}}\n
	statement_in_while_and_if_body__iteration = statement_in_while_and_if_body , statement_in_while_and_if_body__iteration ε;	statement_in_while_and_if_body__iteration → statement_in_while_and_if_body statement_in_while_and_if_body__iteration statement_in_while_and_if_body__iteration → ε	statement_in_while_and_if_body__iteration(1: " ", "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "WHILE", "GET", "PUT", ",", "CONTINUE", ",") → statement_in_while_and_if_body statement_in_while_and_if_body__iteration statement_in_while_and_if_body__iteration(1: !" ", !"(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !" +", !" -, !"IF", !"WHILE", !"GET", !"PUT", !", !"CONTINUE", !"BREAK") → ε		statement_in_while_and_if_body__iteration = statement_in_while_and_if_body >> statement_in_while_and_if_body__iteration "";	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0, T_CONTINUE_WHILE_0, T_EXIT_WHILE_0 }, { "statement_in_while_and_if_body__iteration", {\n {LA_IS, {"" }, 2, { "statement_in_while_and_if_body", "statement_in_while_and_if_body__iteration" }}\n }}}}\n {LA_NOT, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0, T_CONTINUE_WHILE_0, T_EXIT_WHILE_0 }, { "statement_in_while_and_if_body__iteration", {\n {LA_IS, {"" }, 0, { "" }}\n }}}}\n
while_cycle_head_expression = expression;	while_cycle_head_expression = expression;	while_cycle_head_expression → expression	while_cycle_head_expression(1: "(", "NOT", "+", "-", " ", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	while_cycle_head_expression = SAME_RULE(expression);	while_cycle_head_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "while_cycle_head_expression", {\n {LA_IS, {"" }, 1, { "expression" }}\n }}}}\n
while_cycle = "WHILE" , while_cycle_head_expression , block_statements_in_while_and_if_body;	while_cycle = "WHILE", while_cycle_head_expression , block_statements_in_while_and_if_body;	while_cycle → "WHILE" while_cycle_head_expression block_statements_in_while_and_if_body	while_cycle(1: "WHILE") → "WHILE" while_cycle_head_expression block_statements_in_while_and_if_body	while_cycle = tokenWHILE >> while_cycle_head_expression >> block_statements_in_while_and_if_body;	while_cycle = tokenWHILE >> while_cycle_head_expression >> block_statements_in_while_and_if_body;	{LA_IS, {T_WHILE_0 }, { "while_cycle", {\n {LA_IS, {"" }, 3, { T_WHILE_0, "while_cycle_head_expression", "block_statements_in_while_and_if_body" }}\n }}}}\n
	statements__or_block_statements = statement__iteration block_statements;	statements__or_block_statements → statement__iteration statements__or_block_statements → block_statements	statements__or_block_statements(1: " ", "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "WHILE", "GET", "PUT", ",") → statement__iteration statements__or_block_statements(1: "(") → block_statements		statements__or_block_statements = statement__iteration block_statements;	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statements__or_block_statements", {\n {LA_IS, {"" }, 1, { "statement__iteration" }}\n }}}}\n {LA_IS, {T_BEGIN_BLOCK_0 }, { "statements__or_block_statements", {\n {LA_IS, {"" }, 1, { "block_statements" }}\n }}}}\n
input_rule = "GET" , (ident , [index_action] "(" , ident , [index_action] , ") ");	input_rule = "GET", argument_for_input;	input_rule → "GET" argument_for_input	input_rule(1: "GET") → "GET" argument_for_input	input_rule = tokenGET >> (ident >> -index_action tokenGROUPEXPRESSIONBEGIN >> ident >> -index_action >> tokenGROUPEXPRESSIONEND);	input_rule = tokenGET >> argument_for_input;	{LA_IS, {T_INPUT_0 }, { "input_rule", {\n {LA_IS, {"" }, 2, { T_INPUT_0, "argument_for_input" }}\n }}}}\n
	argument_for_input = ident , index_action__optional; argument_for_input = "(" , "ident", "index_action__optional", ")";	argument_for_input → ident index_action__optional argument_for_input → "(" "ident" "index_action__optional" ")"	argument_for_input(1: " ") → ident index_action__optional argument_for_input(1: "(") → "(" "ident" "index_action__optional" ")"		argument_for_input = ident >> index_action__optional tokenGROUPEXPRESSIONBEGIN >> ident >> index_action__optional >> tokenGROUPEXPRESSIONEND;	{LA_IS, { " ident_terminal " }, { "argument_for_input", {\n {LA_IS, {"" }, 2, { "ident", "index_action__optional" }}\n }}}}\n {LA_IS, { "(", { "argument_for_input", {\n {LA_IS, {"" }, 4, { "(", "ident", "index_action__optional", ")" }}\n }}}}\n
output_rule = "PUT" , expression;	output_rule = "PUT", expression;	output_rule → "PUT" expression	output(1: "PUT") → "PUT" expression	output_rule = tokenPUT >> expression;	output_rule = tokenPUT >> expression;	{LA_IS, {T_OUTPUT_0 }, { "output_rule", {\n {LA_IS, {"" }, 2, {T_OUTPUT_0, "expression" }}\n }}}}\n
statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule " ,"; statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule " ,";	statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule " ,";	statement → expression_or_cond_block__with_optional_assign statement → while_cycle statement → input_rule statement → output_rule statement → " ,"	statement(1: " ", " ", 2: !"") → expression_or_cond_block__with_optional_assign statement(1: "(", "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression_or_cond_block__with_optional_assign statement(1: "WHILE") → while_cycle statement(1: "GET") → input_rule	statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule tokenSEMICOLON;	statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule tokenSEMICOLON;	{LA_IS, { "(", T_NOT_0, " ident_terminal ", "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0 }, { "statement", {\n {LA_IS, {"" }, 1, {"expression_or_cond_block__with_optional_assign"}}\n }}}}\n {LA_IS, {T_WHILE_0 }, { "statement", {\n

	"Y" "Z";	letter_in_lower_case → "d" letter_in_lower_case → "e" letter_in_lower_case → "f" letter_in_lower_case → "g" letter_in_lower_case → "h" letter_in_lower_case → "i" letter_in_lower_case → "j" letter_in_lower_case → "k" letter_in_lower_case → "l" letter_in_lower_case → "m" letter_in_lower_case → "n" letter_in_lower_case → "o" letter_in_lower_case → "p" letter_in_lower_case → "q" letter_in_lower_case → "r" letter_in_lower_case → "s" letter_in_lower_case → "t" letter_in_lower_case → "u" letter_in_lower_case → "v" letter_in_lower_case → "w" letter_in_lower_case → "x" letter_in_lower_case → "y" letter_in_lower_case → "z"	letter_in_lower_case(1: "d") → "d" letter_in_lower_case(1: "e") → "e" letter_in_lower_case(1: "f") → "f" letter_in_lower_case(1: "g") → "g" letter_in_lower_case(1: "h") → "h" letter_in_lower_case(1: "i") → "i" letter_in_lower_case(1: "j") → "j" letter_in_lower_case(1: "k") → "k" letter_in_lower_case(1: "l") → "l" letter_in_lower_case(1: "m") → "m" letter_in_lower_case(1: "n") → "n" letter_in_lower_case(1: "o") → "o" letter_in_lower_case(1: "p") → "p" letter_in_lower_case(1: "q") → "q" letter_in_lower_case(1: "r") → "r" letter_in_lower_case(1: "s") → "s" letter_in_lower_case(1: "t") → "t" letter_in_lower_case(1: "u") → "u" letter_in_lower_case(1: "v") → "v" letter_in_lower_case(1: "w") → "w" letter_in_lower_case(1: "x") → "x" letter_in_lower_case(1: "y") → "y" letter_in_lower_case(1: "z") → "z"	D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;	D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;		
letter_in_upper_case = "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V" "W" "X" "Y" "Z";	letter_in_upper_case = "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V" "W" "X" "Y" "Z";	letter_in_upper_case → "A" letter_in_upper_case → "B" letter_in_upper_case → "C" letter_in_upper_case → "D" letter_in_upper_case → "E" letter_in_upper_case → "F" letter_in_upper_case → "G" letter_in_upper_case → "H" letter_in_upper_case → "I" letter_in_upper_case → "J" letter_in_upper_case → "K" letter_in_upper_case → "L" letter_in_upper_case → "M" letter_in_upper_case → "N" letter_in_upper_case → "O" letter_in_upper_case → "P" letter_in_upper_case → "Q" letter_in_upper_case → "R" letter_in_upper_case → "S" letter_in_upper_case → "T" letter_in_upper_case → "U" letter_in_upper_case → "V" letter_in_upper_case → "W" letter_in_upper_case → "X" letter_in_upper_case → "Y" letter_in_upper_case → "Z"	letter_in_upper_case(1: "A") → "A" letter_in_upper_case(1: "B") → "B" letter_in_upper_case(1: "C") → "C" letter_in_upper_case(1: "D") → "D" letter_in_upper_case(1: "E") → "E" letter_in_upper_case(1: "F") → "F" letter_in_upper_case(1: "G") → "G" letter_in_upper_case(1: "H") → "H" letter_in_upper_case(1: "I") → "I" letter_in_upper_case(1: "J") → "J" letter_in_upper_case(1: "K") → "K" letter_in_upper_case(1: "L") → "L" letter_in_upper_case(1: "M") → "M" letter_in_upper_case(1: "N") → "N" letter_in_upper_case(1: "O") → "O" letter_in_upper_case(1: "P") → "P" letter_in_upper_case(1: "Q") → "Q" letter_in_upper_case(1: "R") → "R" letter_in_upper_case(1: "S") → "S" letter_in_upper_case(1: "T") → "T" letter_in_upper_case(1: "U") → "U" letter_in_upper_case(1: "V") → "V" letter_in_upper_case(1: "W") → "W" letter_in_upper_case(1: "X") → "X" letter_in_upper_case(1: "Y") → "Y" letter_in_upper_case(1: "Z") → "Z"	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	\\	
				STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) ((!(qi::alpha qi::char_("_")))); BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY); BOUNDARY = BOUNDARY__SPACE BOUNDARY__TAB BOUNDARY__VERTICAL_TAB BOUNDARY__FORM_FEED BOUNDARY__CARRIAGE_RETURN BOUNDARY__LINE_FEED BOUNDARY__NULL; BOUNDARY__SPACE = " "; BOUNDARY__TAB = "\t"; BOUNDARY__VERTICAL_TAB = "\v"; BOUNDARY__FORM_FEED = "\f"; BOUNDARY__CARRIAGE_RETURN = "\r"; BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \ STRICT_BOUNDARIES, \ BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) (!(qi::alpha qi::char_("_"))); BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY); BOUNDARY = BOUNDARY__SPACE BOUNDARY__TAB BOUNDARY__VERTICAL_TAB BOUNDARY__FORM_FEED BOUNDARY__CARRIAGE_RETURN BOUNDARY__LINE_FEED BOUNDARY__NULL; BOUNDARY__SPACE = " "; BOUNDARY__TAB = "\t"; BOUNDARY__VERTICAL_TAB = "\v"; BOUNDARY__FORM_FEED = "\f"; BOUNDARY__CARRIAGE_RETURN = "\r"; BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \ STRICT_BOUNDARIES, \ BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	\\ \\ \\ { LA_IS, {T_NAME_0 }, { "program____part1", { LA_IS, {""}, 7, {T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration_optional", T_SEMICOLON_0 }}} }}}; \ }; \ "program_rule"	
							\\ \\ \\ { LA_IS, {T_NAME_0 }, { "program____part1", { LA_IS, {""}, 7, {T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration_optional", T_SEMICOLON_0 }}} }}}; \ }; \ "program_rule"